📖 Computer Graphics 😎 ⚡🤭 Note 🤫⚡

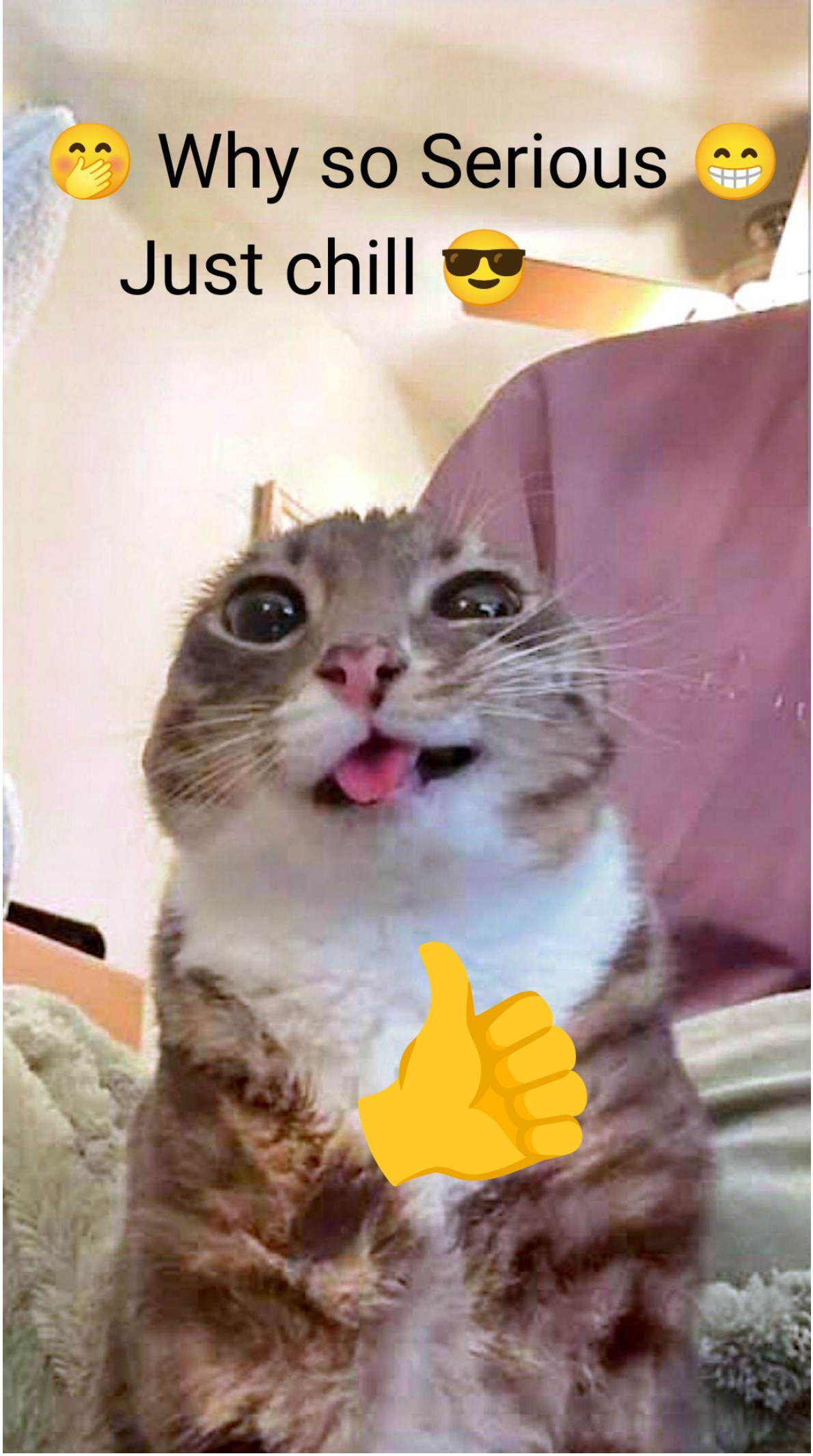
👉 Mid Term (2026) 👈

🔥 Topic 👉 02 : Bresenham's Midpoint Line Algorithm



Created By 😎
Md. Ashikur Rahman Mirza

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Computer Graphics

Mid term

MCQ ⟶ 20 × 1.5 = 30 mark

Obj ⟶ DDA / Bresenham's derivation = 15 mark

code ⟶ 10 mark

theory ⟶ 45 mark (Ans 3 out of 4)

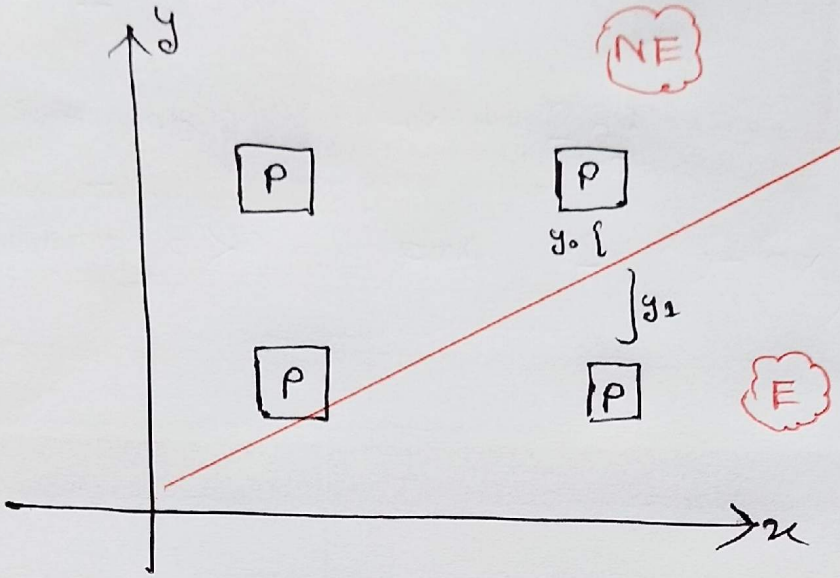
So,
- Slide-1 theory
- RGB, CMYK math + theory
- DDA
- Bresenham's

Bresenham's Mid-Point Line Algorithm

- **Type:** Incremental line drawing algorithm.
- **Purpose:** Approximates a straight line between two points.
- **Approach:** Uses decision parameter to choose the next pixel.
- **Efficiency:** Avoids floating-point calculations.
- **Initial Decision Parameter:** Based on midpoint between two candidate pixels.
- **Update:** Decision parameter updated incrementally.
- **Best For:** Lines with gentle slope ($0 \leq m \leq 1$); can be adapted for other slopes.

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Topic → Bresenhams' Mid Line Algorithum



Bresenham's ও DDA থেকে better কারণ এখানে float point আসে না, তবে select কোনটা হবে pixel ?

যেহেতু, Round System নেই, তাই আমরা,

যার ভ্যালু কম তাকে select করি,

এবং সেই process formula মেনেছে।

Initial Step: $d_{start} = 2dy - dx$

$d_E = 2dy$

$d_{NE} = 2(dy - dx)$

এই formula গুলো Ans বের করবো

When,

$d <= 0$

Select → dE

So,

$d_{new} = d_{old} + dE$

$d > 0$

Select → dNE

So

$d_{new} = d_{old} + dNE$

Math করার Step:

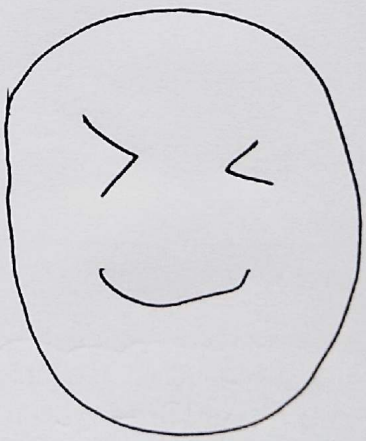
① Intial calculation করবে,

d এর value তুলনা করবো,

if $(d \leq 0)$ তবে dE select করে Next step যাবো

$(d > 0)$ " dNE " " " " " "

② Just same process করবো



Problem1: Using **Bresenham's Midpoint Line Algorithm**, draw a line from point **(2, 3)** to point **(10, 6)**. Show all steps including the initial decision parameter and how it is updated at each step. Plot the pixels selected by the algorithm.

➔ **Is given,**

$X_0 = 02$ $\quad Y_0 = 3$

$X_1 = 10$ $\quad Y_1 = 6$

Step 1: Calculate Differences	**Step 2:** Initial Decision Parameter
$\gg dx = x1 - x0 = 10 - 2 = 8$ $\gg dy = y1 - y0 = 6 - 3 = 3$	***we know,*** $d_{start} = 2.dy - dx$ $= 2 \times 3 - 8$ $= -2 \leq 0 \xrightarrow{E} (3,3)$
Step 3: Find the Next Pixel Values	
3.1 $\therefore d_{new} = d_{old} + dE$ $\therefore d_{new} = -2 + 2dy$ $= -2 + 2 \times 3$ $= 4 > 0 \xrightarrow{NE} (4,4)$	3.2 $\therefore d_{new} = d_{old} + dNE$ $\therefore d_{new} = 4 + 2(dy - dx)$ $= 4 + 2(3 - 8)$ $= -6 \leq 0 \xrightarrow{E} (5,4)$

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3.3	$\therefore d_{new} = d_{old} + dE$ $\therefore d_{new} = -6 + 2dy$ $= -6 + 2 \times 3$ $= 0 \leq 0 \xrightarrow{E} (6,4)$
3.4	$\therefore d_{new} = d_{old} + dE$ $\therefore d_{new} = 0 + 2dy$ $= 0 + 2 \times 3$ $= 6 > 0 \xrightarrow{NE} (7,5)$
3.5	$\therefore d_{new} = d_{old} + dNE$ $\therefore d_{new} = 6 + 2(dy - dx)$ $= 6 + 2 \times (3 - 8)$ $= -4 \leq 0 \xrightarrow{E} (8,5)$
3.6	$\therefore d_{new} = d_{old} + dE$ $\therefore d_{new} = -4 + 2dy$ $= -4 + 2 \times 3$ $= 2 > 0 \xrightarrow{NE} (9,6)$
3.7	$\therefore d_{new} = d_{old} + dNE$ $\therefore d_{new} = 2 + 2(dy - dx)$ $= 2 + 2 \times (3 - 8)$ $= -8 \leq 0 \xrightarrow{E} (10,6)$
Final Pixels Plotted: (2,3), (3,3), (4,4), (5,4), (6,5), (7,5), (8,5), (9,6), (10,6)	

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Problem2: Using **Bresenham's Midpoint Line Algorithm**, draw a line from point **(1, 1)** to point **(7, 6)**. Show all steps including the initial decision parameter and how it is updated at each step. Plot the pixels selected by the algorithm.

➔ **Is given,**

$X_0 = 1$ $\quad Y_0 = 1$

$X_1 = 7$ $\quad Y_1 = 6$

Step 1: Calculate Differences	**Step 2:** Initial Decision Parameter
$\gg dx = x1 - x0 = 7 - 1 = 6$ $\gg dy = y1 - y0 = 6 - 1 = 5$ $\gg dy - dx = 5 - 6 = -1$ $\gg 2(dy - dx) = 2(-1) = -2$	***we know,*** $d_{start} = 2.dy - dx$ $= 2 \times 5 - 6$ $= 4 > 0 \xrightarrow{NE} (2,2)$
Step 3: Find the Next Pixel Values	
3.1 $\therefore d_{new} = d_{old} + dNE$ $\therefore d_{new} = 4 + 2(dy - dx)$ $= 4 - 2$ $= 2 > 0 \xrightarrow{NE} (3,3)$	3.2 $\therefore d_{new} = d_{old} + dNE$ $\therefore d_{new} = 2 + 2(dy - dx)$ $= 2 - 2$ $= 0 \leq 0 \xrightarrow{E} (4,3)$

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3.3	$\therefore d_{new} = d_{old} + dE$ $\therefore d_{new} = 0 + 2dy$ $= 0 + 2 \times 5$ $= 10 > 0 \xrightarrow{NE} (5,4)$	3.4	$\therefore d_{new} = d_{old} + dNE$ $\therefore d_{new} = 10 + 2(dy - dx)$ $= 10 - 2$ $= 8 > 0 \xrightarrow{NE} (6,5)$
3.5	$\therefore d_{new} = d_{old} + dNE$ $\therefore d_{new} = 8 + 2(dy - dx)$ $= 8 - 2$ $= 6 > 0 \xrightarrow{NE} (7,6)$		**Final Pixels Plotted:** (1,1), (2,2), (3,3), (4,3), (5,4), (6,5), (7,6)

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Problem3: Using **Bresenham's Midpoint Line Algorithm**, draw a line from point **(2, 1)** to point **(2, 7)**. Show all steps including the initial decision parameter and how it is updated at each step. Plot the pixels selected by the algorithm.

➔ **Is given,**

$X_0 = 2$ $Y_0 = 1$

$X_1 = 2$ $Y_1 = 7$

Step 1: Calculate Differences

$$\gg dx = x1 - x0 = 2 - 2 = 0$$
$$\gg dy = y1 - y0 = 7 - 1 = 6$$

Since $dx = 0$, this is a **vertical line**.

So, Vertical lines aren't handled normally in Bresenham's algorithm because dx = 0, so x doesn't change. The algorithm relies on x-increment to update the decision parameter, which doesn't work for vertical lines. Special handling is required.

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Problem4: Using **Bresenham's Midpoint Line Algorithm**, draw a line from point **(9, 5)** to point **(15, 13)**. Show all steps including the initial decision parameter and how it is updated at each step. Plot the pixels selected by the algorithm.

➔ **Is given,**

$X_0 = 09$ $Y_0 = 05$

$X_1 = 15$ $Y_1 = 13$

Step 1: Calculate Differences

» $dx = x1 - x0 = 15 - 9 = 6$
» $dy = y1 - y0 = 13 - 5 = 8$
» $dy - dx = 8 - 6 = 2$
» $2(dy - dx) = 2(2) = 4$

Step 2: Initial Decision Parameter

we know,

$$d_{start} = 2.dy - dx$$
$$= 2 \times 8 - 6$$
$$= 10 > 0 \xrightarrow{NE} (10,6)$$

Step 3: Find the Next Pixel Values

3.1

$$\therefore d_{new} = d_{old} + dNE$$
$$\therefore d_{new} = 10 + 2(dy - dx)$$
$$= 10 + 4$$
$$= 14 > 0 \xrightarrow{NE} (11,7)$$

3.2

$$\therefore d_{new} = d_{old} + dNE$$
$$\therefore d_{new} = 14 + 2(dy - dx)$$
$$= 14 + 4$$
$$= 18 > 0 \xrightarrow{NE} (12,8)$$

3.3

$$\therefore d_{new} = d_{old} + dNE$$
$$\therefore d_{new} = 18 + 2(dy - dx)$$
$$= 18 + 4$$
$$= 22 > 0 \xrightarrow{NE} (13,9)$$

3.4

$$\therefore d_{new} = d_{old} + dNE$$
$$\therefore d_{new} = 22 + 2(dy - dx)$$
$$= 22 + 4$$
$$= 26 > 0 \xrightarrow{NE} (14,10)$$

3.5

$$\therefore d_{new} = d_{old} + dNE$$
$$\therefore d_{new} = 26 + 2(dy - dx)$$
$$= 26 + 4$$
$$= 30 > 0 \xrightarrow{NE} (15,11)$$

Final Pixels Plotted:
(9,5), (10,6), (11,7), (12,8), (13,9), (14,10), (15,11)

This is the output of Bresenham's algorithm for this line. The algorithm ensures that the line is as close as possible to the true line while using only integer arithmetic.

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Bresenham's Mid Point Line Algorithm

Slop(m) < 1

$d_{start} = 2d_y - d_x$

when d>0 ----- select NE

$d_{new} = d_{old} + 2(d_y - d_x)$

$x_{i+1} = x_i + 1$

$y_{i+1} = y_i + 1$

when d<=0 ----- select E

$d_{new} = d_{old} + 2d_y$

$x_{i+1} = x_i + 1$

$y_{i+1} = y_i$

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Bresenham's Mid Point Line Algorithm

Slop(m) > 1

$d_{start} = 2d_x - d_y$

when d>0 ----- select NE

$d_{new} = d_{old} + 2(d_x - d_y)$

$x_{i+1} = x_i + 1$

$y_{i+1} = y_i + 1$

when d<=0 ----- select N

$d_{new} = d_{old} + 2d_x$

$x_{i+1} = x_i$

$y_{i+1} = y_i + 1$

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১. $m < 1$ (প্রথম ছবি - $d_{start} = 2d_y - d_x$)

যখন লাইনের ঢাল ১-এর চেয়ে কম হয়, তখন লাইনটি ভূমি বা x-অক্ষের দিকে বেশি ঝুঁকে থাকে।

- **Driving Axis:** এখানে x-অক্ষ হলো ড্রাইভিং অ্যাক্সিস। অর্থাৎ, আমরা প্রতি ধাপে x এর মান ১ করে বাড়াবো ($x_{i+1} = x_i + 1$)।
- **সিদ্ধান্ত:** আমরা দেখবো তার সাথে y এর মান ১ বাড়বে কি বাড়বে না। যদি $d > 0$ হয়, তবে উত্তর-পূর্ব (NE) পিক্সেল সিলেক্ট হবে, আর $d \leq 0$ হলে শুধু পূর্ব (E) পিক্সেল সিলেক্ট হবে।

২. $m > 1$ (দ্বিতীয় ছবি - $d_{start} = 2d_x - d_y$)

যখন ঢাল ১-এর চেয়ে বেশি হয়, তখন লাইনটি উলম্ব বা y-অক্ষের দিকে বেশি খাড়া থাকে।

- **Driving Axis:** এক্ষেত্রে ড্রাইভিং অ্যাক্সিস বদলে গিয়ে y-অক্ষ হয়ে যায়। আমরা প্রতি ধাপে y এর মান ১ করে বাড়াবো ($y_{i+1} = y_i + 1$)।
- **সিদ্ধান্ত:** এখানে আমরা দেখবো তার সাথে x এর মান বাড়বে কি না। যদি $d > 0$ হয়, তবে উত্তর-পূর্ব (NE) পিক্সেল সিলেক্ট হবে, আর $d \leq 0$ হলে শুধু উত্তর (N) পিক্সেল সিলেক্ট হবে।

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Sub.: .....

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(3) Derive Midpoint Line Algorithm to draw a stright line between two points, $P_0(x_0, y_0)$ and $P_n(x_1, y_1)$ on a raster display using only integer arithmetic. Clearly show each step along with the obtainted formulas.

Ans: Here DDA will not accepted becaus here mention integer arithmetic and we know Bresenham midpoint line algorithm follows integer arithmatic formulation.

Derive the Bresenham Mid point Line Algorithms. [From slide]

Sub.: .....

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(5) Find the first four iterations that will be obtained using DDA for points P(2,4) and Q(8,12). Derive the mathmetical equation of a Midpoint Line Algorithm to decide whether to go East or North East to choose the next point.

Ans: Part-1:

Here, we did not use DDA for derivation. Because, here we need to decide the decision pattern. So for derivation part we will use Brezenham's mid point Line Algorithm.

For derivation follow the slides.

Sub.:

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Part-2:

$$m = \frac{12-4}{8-2} = \frac{8}{6} = 1.3$$

$m > 1$ so, Δy will increase by 1. [Line direction "+"]

Now, $m = \frac{\Delta y}{\Delta x}$

$$\Rightarrow \Delta x = \frac{\Delta y}{m}$$

$$= \frac{1}{1.33}$$

$$= .25$$

So, Now formulation will be,

$x = x + .25$ and $y = y + 1$

① $x = 2$ (initial)
$y = 4$
$(2, 4)$

② $x = 2 + .75 = 2.75$
$= 3$
$y = 4 + 1 = 5$
$(3, 5)$

③ $x = 2.75 + .75 = 3.5$
$= 4$
$y = 5 + 1 = 6$
$(4, 6)$

④ $x = 3.5 + .25 = 4.3 = 4$
$y = 7$
$(4, 7)$

⑤ $x = 4.3 + .75 = 5$
$y = 8$
$(5, 8)$

Topic: Derivation

Bresenham mid-point Line

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Derivation ⟹ Breshemhans Mid point

we know, $y = mx + B$ for $m < 1$

$$f(x,y) = ax + by + c$$

now, $y = mx + B$

$$\Rightarrow y = \frac{\Delta y}{\Delta x} x + B$$

$$\Rightarrow (\Delta x) y = (\Delta y) x + B(\Delta x)$$ [উভয় পাশে (Δx) দ্বারা গুণ]

$$\Rightarrow (\Delta y) x - (\Delta x) y + B(\Delta x) = 0$$

now with comparing with $ax + by + c = 0$,

So, $a = \Delta y$, $b = -\Delta x$, $c = (\Delta x) B$

Now,

$$d_{start} = f\left(x_0 + 1, y_0 + \frac{1}{2}\right)$$

$$= a\left(x_0 + 1\right) + b\left(y_0 + \frac{1}{2}\right) + c$$

$$= ax_0 + a + by_0 + \frac{b}{2} + c$$

$$= ax_0 + by_0 + c + a + \frac{b}{2}$$

$$= 0 + a + \frac{b}{2}$$ [$\because ax_0 + by_0 + c = 0$]

So, $d_{start} = a + \frac{b}{2}$

$\Rightarrow 2(d_{start}) = 2a + b$

$\Rightarrow 2(d_{start}) = 2dy + (-dx)$

$\Rightarrow 2(d_{start}) = 2dy - dx$

now, if $d > 0$

then, $\frac{d}{2} > 0$ and $2d > 0$ is correct,

So we can write,

$$d_{start} = 2dy - dx.$$

if, E ($d <= 0$) is chossen,

We know, $d_{new} = f(x+2, y_0 + \frac{1}{2})$

So, $\Delta d = d_{new} - d_{old}$

$= f(x_0+2, y_0 + \frac{1}{2}) - d(x_0+1, y + \frac{1}{2})$

$= a(x_0+2) + b(y_0 + \frac{1}{2}) + c - \left(a(x_0+1) + b(y_0+\frac{1}{2}) + c\right)$

$= a(x_0+2) - a(x_0+1)$

$= ax_0 + 2a - ax_0 - a = a$

now balancing Δd_E by multiply by 2 due to

$2 d_{start} \approx d_{start}$

so, $\Delta d = a$

Now, $\Delta d_E = 2a = 2dy$

if $(d > 0)$ so, NE is chossen :

$$d_{new} = f\left(x_0 + 2, \; y_0 + \frac{3}{2}\right)$$

so, $\Delta d = d_{new} - d_{old}$

$$= f\left(x_0 + 2, \; y_0 + \frac{3}{2}\right) - f\left(x_0 + 1, \; y_0 + \frac{1}{2}\right)$$

$$= a(x_0 + 2) + b\left(y_0 + \frac{3}{2}\right) + c - \left(a(x_0 + 1) + b\left(y_0 + \frac{1}{2}\right) + c\right)$$

$$= ax_0 + 2a + by_0 + \frac{3b}{2} - ax_0 - a - by_0 - \frac{b}{2}$$

$$= 2a - a + \frac{3b}{2} - \frac{b}{2}$$

$$= a + b$$

now adjusting Δd_{ne} by multiply by 2

so, $\Delta d_{ne} = 2(a+b)$

$$= 2(dy - dx)$$

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So,

$$d_{start} = 2dy - dx$$

d_E, when $(d \leq 0)$, $d_E = 2dy$

d_{nE}, when $(d > 0)$, $d_{nE} = 2(dy - dx)$



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